

Set 22: Molar mass

Notes

Set 22: Exercises

1. Calculate the molar masses of
 - (a) potassium hydroxide, KOH
 - (b) copper(II) chloride, CuCl_2
 - (c) aluminium chloride, AlCl_3
 - (d) calcium hydroxide, Ca(OH)_2
 - (e) ammonium oxalate, $(\text{NH}_4)_2\text{C}_2\text{O}_4$
 - (f) sodium carbonate-10-water, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
 - (g) tungsten carbide, WC
 - (h) methane, CH_4
 - (i) sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
 - (j) silver chloride
 - (k) zinc iodide
 - (l) sodium sulfate
 - (m) iron(III) sulfate
 - (n) sulfur dioxide
 - (o) sulfuric acid
2. Carbon dating is possible due to the composition of the C-14 isotope in living organisms.
Find out:
 - (a) how a C-14 atom is similar and different to a C-12 atom;
 - (b) any other isotopes of carbon;
 - (c) which isotope of carbon is the most abundant and how you can verify this;
 - (d) how carbon dating works.
3. Hydrogen has three isotopes; hydrogen, deuterium and tritium.
 - (a) Describe the composition of each of these atoms.
 - (b) Where are deuterium and tritium used?